Manjit Chakraborty

Electronics & Communication Engineer

Github | Linkdin | Portfolio

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Agartala, Tripura

Electronics & Communication engineer with 3 years in UAV technology, specializing in embedded systems, PCB design, 3D design, computer vision, and machine learning. Proficient in C, C++, and Python, with hands-on experience in I2C, SPI, UART, and CAN protocols and strong hardware-software integration. Trained 200+ students in UAV and drone systems.

Education

B.Tech in Electronics & Communication 2023-2027

Tripura Institute of Technology Narsingarh,Tripura

Experience

Executive Member - Drone Technology Center, Tripura Institute of Technology Sep 2023 - Present

- Designed and developed a UAV system that ranked AIR 12 in National Innovation Challenge for Drone Application and Research, organized by the Drone Federation of India (certificate↗)
- Selected as State Ambassador, Drone Federation of India. (appointment letter↗)
- Designed and fabricated a custom flight controller that won 1st place at the Northeast Techathon (IIM Calcutta) & got 1 lakh Fund Under NGIN Scheme. (certificate↗)
- Trained and mentored students and Indian Army personnel of 6th Battalion Sikh Regiment.(certificate↗)

AIML Intern - IIT Delhi Jul 2026 - Aug 2026

- Trained a gesture-recognition model to control a drone in real time, with no heavy onboard hardware, achieving 94% accuracy at ~40 ms latency.
- Integrated the model with the drone flight stack over MAVLink, translating 6 hand gestures into live flight commands.
- Applied ML methods (CNN, SVM, Random Forest) using Python libraries including TensorFlow/PyTorch, OpenCV, MediaPipe, scikit-learn, NumPy, and Pandas.
- Built and evaluated the pipeline on a dataset of 7890 samples, improving accuracy from 82% to 94% through tuning and augmentation. (github↗)

Projects

Secure Bootloader with OTA Firmware Update (STM32F103)

- Designed and implemented a custom bootloader in C for the STM32 Cortex-M3 with a software A/B flash layout, page-by-page image swap, and resumable updates that survive power loss during flashing.
- Added firmware integrity and authenticity checks using CRC32, SHA-256, and Ed25519 signatures, plus an anti-downgrade version counter, so corrupted, unsigned, or older images are rejected before they run.
- Built automatic rollback with a boot-confirmation flag and watchdog, and a Python host tool for signing and sending images over UART; validated with fault-injection tests including power cuts, corrupted bytes, bad signatures, and crashing firmware.

Function Generator - Github↗

- Designed and built an analog function generator using a 555 timer in astable mode and a 741 op-amp, producing square, triangle, and sine waveforms.
- Generated the square wave with the 555 timer and converted it to triangle and sine outputs using op-amp integrator and wave-shaping (RC filter) stages, with adjustable frequency through a potentiometer.
- Tested and debugged the circuit on a breadboard using an oscilloscope, measuring frequency and amplitude across [100 Hz to 3.1 kHz].

Achievements & Certifications

- 1st Prize, Robo Fight Competition, ICFAI University, Tripura (certificate↗) **2026**
- 1st Place, EUREKA Pitching Competition, E-Cell, NFSU Tripura Campus (certificate↗) **2026**
- 2nd Position, Drone Race Competition, TCEA (certificate↗) **2026**
- 3rd Position, Sky Rush Drone Competition, TIT (certificate↗) **2026**
- RTL, IP Integration and SoC Signoff, NIELIT, MeitY, Government of India (certificate↗) **2026**
- 2nd Position, STRATATHON 1.0 Inter-College Business Plan Competition, IIC, TCEA (certificate↗) **2025**
- Finalist, Yukti Innovation Challenge, Institution's Innovation Council (certificate↗) **2025**
- Android Development, NIELIT (certificate↗) **2024**
- AI for Beginners, HP LIFE, HP Foundation (certificate↗) **2024**

Skills

- **Programming Languages:** C, C++, Python,
- **Hardware & Protocols:** ARM, STM32F4, Arduino, ESP32, Raspberry Pi, GPIO, ADC/DAC, PWM, I^2C, UART, SPI, CAN, MAVLink,
- **Frameworks :** TensorFlow, PyTorch, scikit-learn, OpenCV, YOLO, ArduPilot, Betaflight, Pixhawk, Mission Planner, QGroundControl
- **Tools & Platforms:** MATLAB, STM32CubeIDE, Proteus, SolidWorks, Fusion 360, Intel Quartus II , EasyEDA, KiCad, LTspice, Git, GitHub